

Ex. No 1: Evidence Acquisition Using AccessData FTK Imager

Course / Lab: Digital Forensics Laboratory

Experiment: Evidence Acquisition Using AccessData FTK Imager[cite: 2]

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Overview

Forensic Toolkit (FTK) Imager is an industry-standard computer forensics software product used for acquiring and analyzing digital evidence securely[cite: 2]. This experiment demonstrates the methodology for acquiring both volatile memory (RAM) and non-volatile storage (logical partitions) while preserving evidence integrity through cryptographic hashing.

Experiment Objectives

1. Capture volatile memory and system states from a live environment.
 2. Acquire non-volatile evidence via logical drive/partition imaging to optimize laboratory time[cite: 3].
 3. Verify data integrity by computing and validating MD5 and SHA1 checksums[cite: 3].
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Part 1: Acquiring Volatile Memory (RAM)

Volatile memory captures running processes, active network sockets, and temporary data.

Steps:

1. Launch the memory acquisition utility (e.g., WinPmem/FTK memory capture module) with administrator privileges[cite: 3].
2. Define the output destination file (`.raw` / `.mem`) and monitor the live memory dumping progress[cite: 3].
3. Allow the driver to complete the memory transfer and verify driver unloading.

Capturing Memory in Progress

```
C:\> Administrator: Command Prompt - winpmem_mini_x64_rc2.exe lnba.raw

Padding from 0x5CD1E000 to 0x617FF000
pad
- length: 0x4ae1000

07% 0x5CD1E000 .....
copy_memory
- start: 0x617ff000
- end: 0x61800000

08% 0x617FF000 .
Padding from 0x61800000 to 0x100000000
pad
- length: 0x9e800000

08% 0x61800000 .....
08% 0x61800000 .....
08% 0x61800000 .....
08% 0x61800000 .....
copy_memory
- start: 0x100000000
- end: 0x48f800000

21% 0x100000000 .....XXXXX..X.....
26% 0x132000000 .....
30% 0x164000000 .....
34% 0x196000000 .....X.....
39% 0x1C8000000 .....X.....
43% 0x1FA000000 .....
47% 0x22C000000 .....
51% 0x25E000000 .....
```

Figure 1.1: Live memory dumping execution.

Part 2: Acquiring Non-Volatile Memory (Logical Drive / Partition)

To streamline forensic acquisition workflows, a specific partition (F:\) is targeted instead of the entire physical disk.

Steps:

1. Open FTK Imager and initiate the **Create Disk Image** wizard[cite: 3].
2. Select **Logical Drive** as the source type and choose the target partition (F:\)[cite: 3].
3. Fill in the **Evidence Item Information** (Case Number, Evidence Number, Unique Description, Examiner, and Notes)[cite: 3].
4. Configure the destination folder (E:\), image filename (EXP), fragment size, and enable **Verify images after they are created**[cite: 3].
5. Click **Start** to begin imaging[cite: 3].

Evidence Item Information	Image Source & Destination Setup	Configuring Destination & Fragment Size
[image]	[image]	[image]
Figure 2.1: Case and examiner metadata.	Figure 2.2: Source partition selection (F:\).	Figure 2.3: Destination path and options.

🔍 Verification & Cryptographic Hashes

To ensure court admissibility and data integrity, FTK Imager automatically calculates cryptographic hashes upon completion and cross-verifies them against the report hash[cite: 3].

Verification Result

[image]

Figure 3.1: Drive/Image Verify Results showing successful MD5 and SHA1 matches.

Checksum Summary:

Image Type: Raw / Logical Drive Image (Result.001)
Sector Count: 31506432

[MD5 Hash]
Computed hash: 1f8957e55573455d076e144a74afef4
Report hash: 1f8957e55573455d076e144a74afef4
Verify result: Match

[SHA1 Hash]
Computed hash: 613f621a072747853feb89ed81700c749c2cc7b1
Report hash: 613f621a072747853feb89ed81700c749c2cc7b1
Verify result: Match

[Bad Blocks List]
Bad block(s) in image: No bad blocks found in image